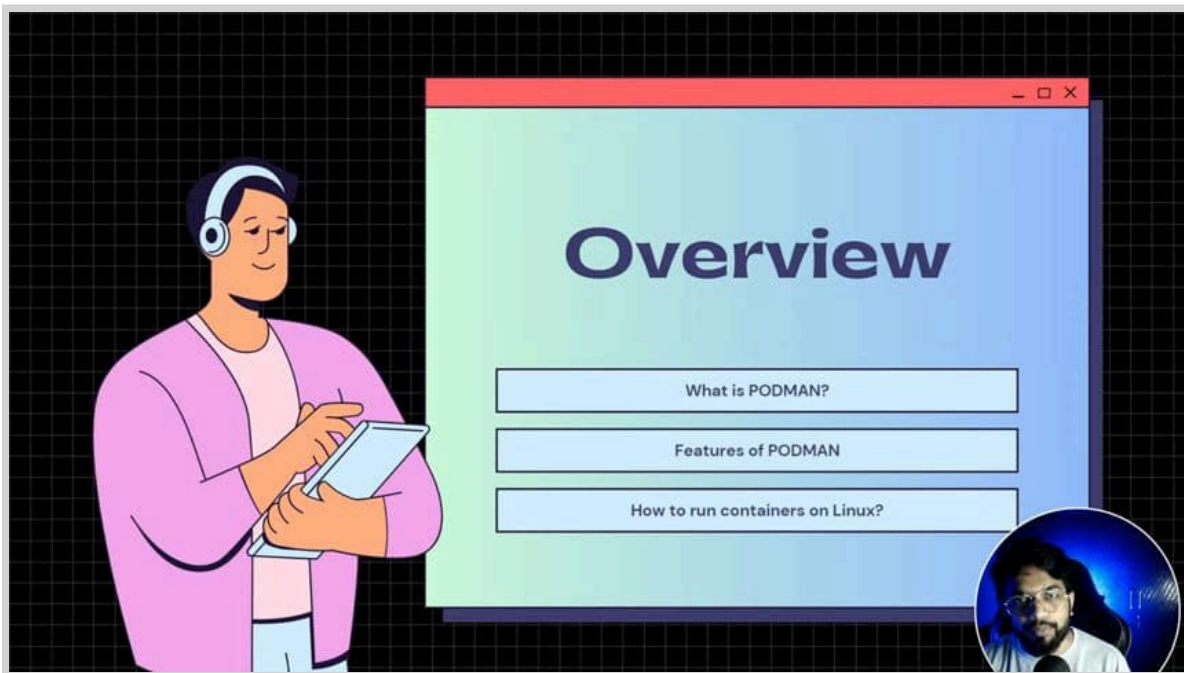


00:49



01:22



Podman is opensource containerzation tool.

01:35

What is a Podman?

- **Podman** provides a secure and lightweight alternative to Docker for creating and managing containers on Linux systems.
- **Podman** allows you to package an application and its dependencies into a single container that can run anywhere, without any dependency on the host system.



[02:14](#)



[02:29](#)

Features of Podman

- **Daemonless:** Podman does not require background daemon to run, make it more light weight and efficient than docker.
- **Rootless Containers:** To run Podman containers, root privileges not required.

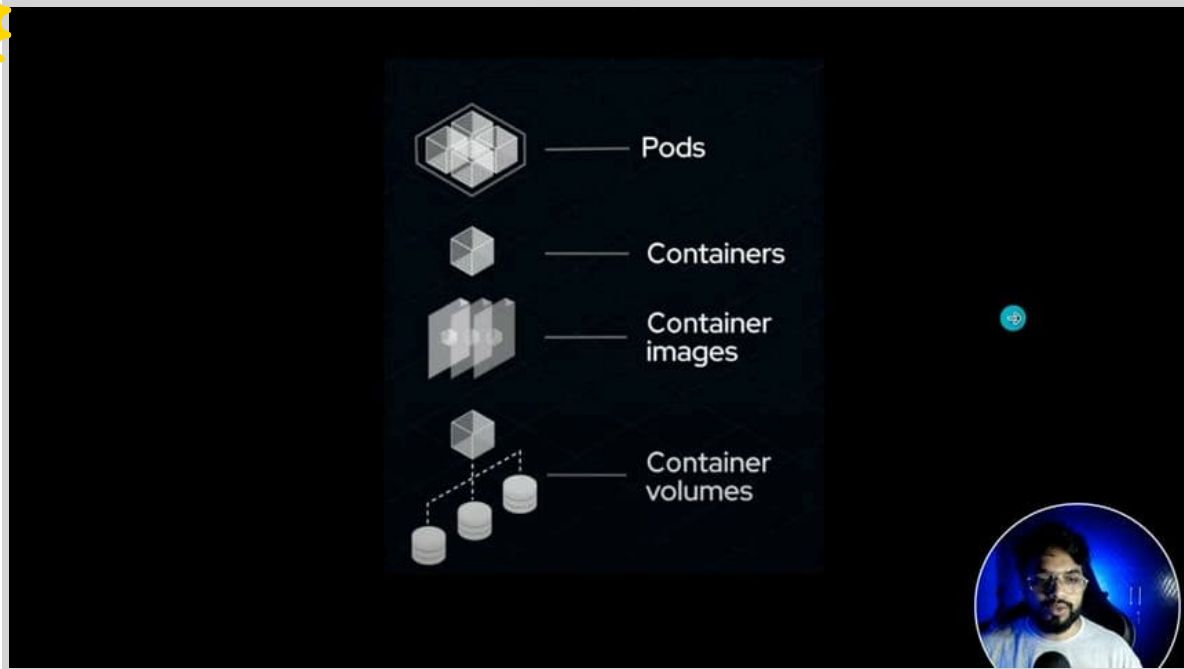


[03:40](#)

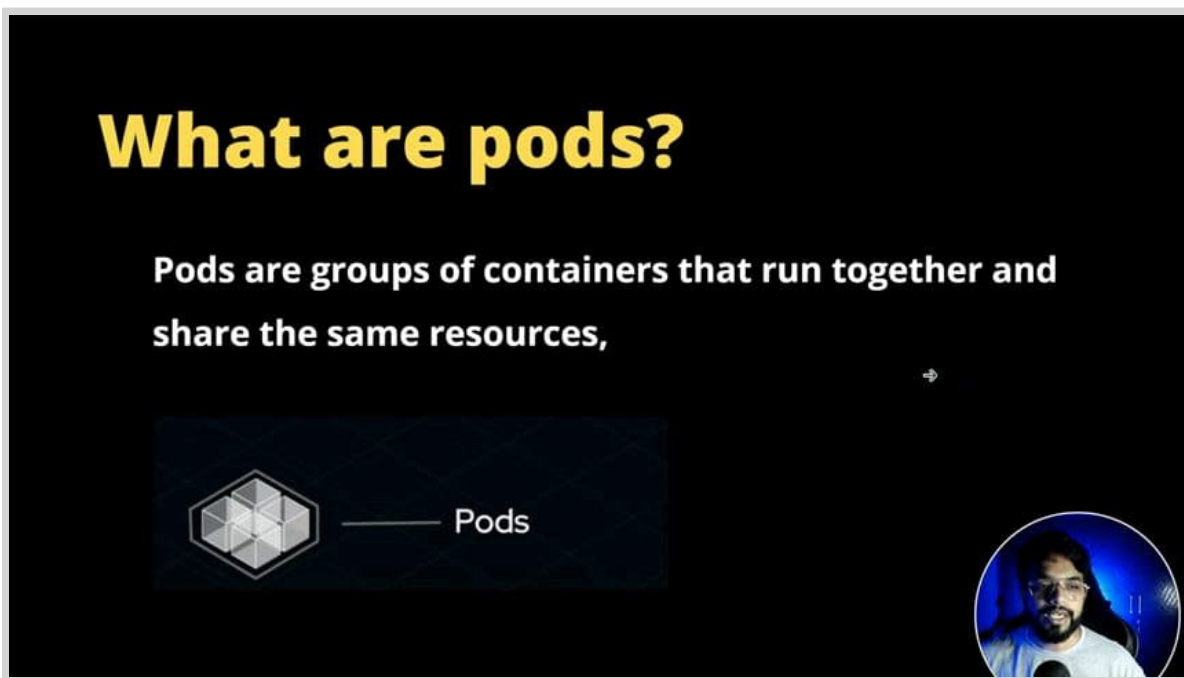
- **Multi-container Applications:** Podman supports running multiple containers as part of an application, and it provides tools for managing and connecting these containers.
- **More Secure:** Podman launches each container with a security-enhanced linux (SELinux) label and it's rootless feature make it more secure



[04:39](#)



[04:47](#)



[05:43](#)

Users can run Podman on various Linux distributions, such as

- Red Hat Enterprise Linux,
- Fedora,
- CentOS,
- Ubuntu.



06:03

Buildah and Skopeo

Both used with Podman to manager container images and build container.

Buildah: Used to create container images

Skopeo: Used to push and pull images from remote registries.



06:47

Steps for installation

```
yum/dnf install podman -y
```



[08:27](#)

Creating Alias

```
alias docker=podman
```

To Check Podman Version

```
podman -v
```



[09:17](#)

```
[root@cs-client ~]#  
[root@cs-client ~]# alias docke=podman  
[root@cs-client ~]#  
[root@cs-client ~]# docker -v  
podman version 4.3.1  
[root@cs-client ~]#  
[root@cs-client ~]#
```



09:33

Checking Podman Registries 🌐

podman info

To Search for a image in registry

podman search nginx



11:46

Checking already downloaded images

`podman images`

To download an image

`podman pull docker.io/library/nginx`



[12:47](#)

```
[root@cs-client ~]#  
[root@cs-client ~]# docker images  
REPOSITORY TAG IMAGE ID CREATED SIZE  
[root@cs-client ~]#  
[root@cs-client ~]#  
[root@cs-client ~]# docker pull nginx  
? Please select an image:  
  > registry.access.redhat.com/nginx:latest  
    registry.redhat.io/nginx:latest  
    docker.io/library/nginx:latest
```



above pic you can use arrow key to up and down

[13:53](#)

```
[root@cs-client ~]#  
[root@cs-client ~]#  
[root@cs-client ~]# docker pull nginx  
✓ docker.io/library/nginx:latest  
Trying to pull docker.io/library/nginx:latest...  
Getting image source signatures  
Copying blob 75576236abf5 done  
Copying blob 26c5c85e47da done  
Copying blob 4f3256bdf66b done  
Copying blob 2019c71d5655 done  
Copying blob 8c767bdb9ae done  
Copying blob 78e14bb05fd3 done  
Copying config 6efc10a051 done  
Writing manifest to image destination  
Storing signatures  
6efc10a0510f143a90b69dc564a914574973223e88418d65c1f8809e08dc0a1f  
[root@cs-client ~]#
```



[14:25](#)

To run an image

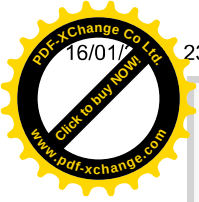
```
podman run docker.io/library/nginx
```

To see the running containers

```
podman ps
```



[15:30](#)



```
[root@cs-client ~]#  
[root@cs-client ~]# docker run -d docker.io/library/nginx  
85a06308da03a8022ea3a62b5cdfdcae802ed4fa6cd97c5b9ccfa84c3919ae3e  
[root@cs-client ~]#
```



[16:21](#)

```
[root@cs-client ~]#  
[root@cs-client ~]# docker run -d docker.io/library/nginx  
85a06308da03a8022ea3a62b5cdfdcae802ed4fa6cd97c5b9ccfa84c3919ae3e  
[root@cs-client ~]#  
[root@cs-client ~]# docker ps  
CONTAINER ID   IMAGE                                COMMAND                  CR  
EATED          STATUS          PORTS                NAMES  
85a06308da03   docker.io/library/nginx:latest      nginx -g daemon o...    31  
seconds ago    Up 31 seconds ago                    hungry_bhaskara  
[root@cs-client ~]#  
[root@cs-client ~]#
```



[16:48](#)

To run an image

```
podman run docker.io/library/nginx
```

To see the running containers

```
podman ps
```



18:06

Port Binding of container

```
podman run -dt -p 8080:80/tcp docker.io/library/nginx
```



meaning of this is bind the nginx 80/tcp port with linux 8080 port

19:34

To Stop a container

```
podman stop <container_name>
```



[19:57](#)

```
[root@cs-client ~]#  
[root@cs-client ~]# docker ps  
CONTAINER ID   IMAGE                                COMMAND                  CR  
EATED          STATUS          PORTS                  NAMES  
85a06308da03   docker.io/library/nginx:latest     nginx -g daemon o...    6  
minutes ago    Up 6 minutes ago                                hungry_  
bhaskara  
76f4c5d785af   docker.io/library/nginx:latest     nginx -g daemon o...    Ab  
out a minute ago Up About a minute ago  0.0.0.0:8080->80/tcp    compass  
ionate_cartwright  
[root@cs-client ~]#  
[root@cs-client ~]# docker stop 85a06308da03  
85a06308da03  
[root@cs-client ~]# docker st
```

×



[20:02](#)

```
[root@cs-client ~]#
[root@cs-client ~]# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CR
EATED         STATUS          PORTS                  NAMES
85a06308da03   docker.io/library/nginx:latest     nginx -g daemon o...    6
minutes ago    Up 6 minutes ago                                hungry_
bhaskara
76f4c5d785af   docker.io/library/nginx:latest     nginx -g daemon o...    Ab
out a minute ago Up About a minute ago 0.0.0.0:8080->80/tcp compass
ionate_cartwright
[root@cs-client ~]#
[root@cs-client ~]# docker stop 85a06308da03
85a06308da03
[root@cs-client ~]# docker stop █
```



20:05

```
[root@cs-client ~]#
[root@cs-client ~]# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CR
EATED         STATUS          PORTS                  NAMES
85a06308da03   docker.io/library/nginx:latest     nginx -g daemon o...    6
minutes ago    Up 6 minutes ago                                hungry_
bhaskara
76f4c5d785af   docker.io/library/nginx:latest     nginx -g daemon o...    Ab
out a minute ago Up About a minute ago 0.0.0.0:8080->80/tcp compass
ionate_cartwright
[root@cs-client ~]#
[root@cs-client ~]# docker stop 85a06308da03
85a06308da03
[root@cs-client ~]# docker stop 76f4c5d785af
76f4c5d785af
[root@cs-client ~]#
```



after the stop there is container id

20:25

To run multiple containers with different port

```
podman run -dt -p 8081:80/tcp docker.io/library/nginx
```

```
podman run -dt -p 8082:80/tcp docker.io/library/nginx
```



[21:22](#)

```
[root@cs-client ~]#  
[root@cs-client ~]# █
```



[21:19](#)

```
[root@cs-client ~]#
[root@cs-client ~]# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
[root@cs-client ~]# docker run -dt -p 8080:80/tcp docker.io/library/nginx
99caec9f7093b5301e3fdf67efc2adc7d7e9c1d1586fbbbed54847a78071cac33
[root@cs-client ~]#
[root@cs-client ~]# docker run -dt -p 8082:80/tcp docker.io/library/nginx
a38f9f79712b295ae40d990e57c7b04b42c113383e614f72095cf0467697d0c0
[root@cs-client ~]#
```

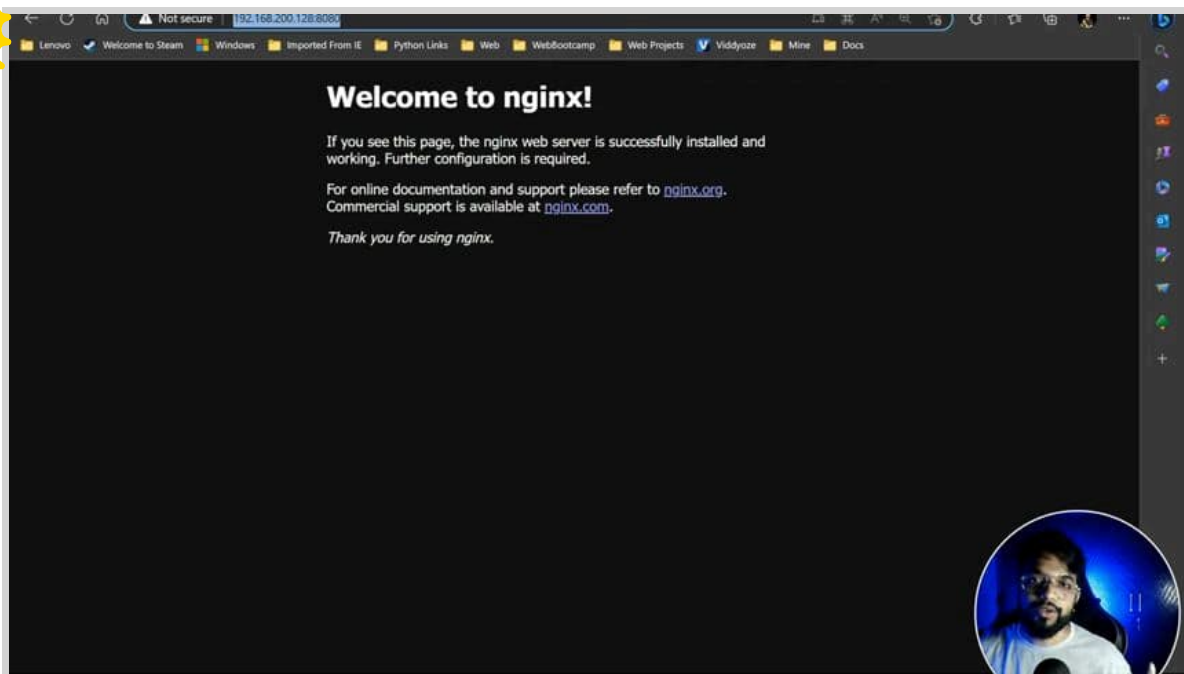


[21:25](#)

```
[root@cs-client ~]#
[root@cs-client ~]# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CR
EATED          STATUS        PORTS        NAMES
99caec9f7093   docker.io/library/nginx:latest      nginx -g daemon o...  14
seconds ago    Up 14 seconds ago    0.0.0.0:8080->80/tcp quizzical_jepse
n
a38f9f79712b   docker.io/library/nginx:latest      nginx -g daemon o...  7
seconds ago    Up 7 seconds ago    0.0.0.0:8082->80/tcp pensive_rosalin
d
[root@cs-client ~]#
```



[21:38](#)



22:23



23:00

```
[root@cs-client ~]#
[root@cs-client ~]# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CR
EATED         STATUS          PORTS                NAMES
99caec9f7093   docker.io/library/nginx:latest      nginx -g daemon o...    5
minutes ago   Up 5 minutes ago  0.0.0.0:8080->80/tcp  quizzical_jepsen
a38f9f79712b   docker.io/library/nginx:latest      nginx -g daemon o...    5
minutes ago   Up 5 minutes ago  0.0.0.0:8082->80/tcp  pensive_rosalind
8392b2df05c3   docker.io/library/nginx:latest      nginx -g daemon o...    4
seconds ago   Up 4 seconds ago                nginx-container1
[root@cs-client ~]#
[root@cs-client ~]# docker stop nginx-container1
nginx-container1
[root@cs-client ~]#
```



[23:11](#)

To see all the existing e to yours

podman ps -a

To remove the existing containers

podman rm <container_name>



[24:34](#)

How to Create our own image?



[25:25](#)

Let's create a basic python app

Create a python file named `app.py` and add following line
`print("Hello World!")`

Run our python program using
`python3 app.py`



[25:34](#)

```
FROM python:3.8
WORKDIR /app
COPY . .
CMD ["python", "app.py"]
```

```
print ("Hey Buddy!")
```

2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2

: wq

```
[root@cs-client myimage]#  
[root@cs-client myimage]# vi app.py  
[root@cs-client myimage]#  
[root@cs-client myimage]# cat app.py  
print ("Hey Buddy!")  
[root@cs-client myimage]# █
```



[26:51](#)

```
[root@cs-client myimage]#  
[root@cs-client myimage]# python3 -V  
Python 3.6.8  
[root@cs-client myimage]# █
```



[27:07](#)

```
[root@cs-client myimage]#  
[root@cs-client myimage]# python3 -V  
Python 3.6.8  
[root@cs-client myimage]#  
[root@cs-client myimage]# python3 app.py  
Hey Buddy!  
[root@cs-client myimage]#
```



[27:35](#)

```
[root@cs-client myimage]#  
[root@cs-client myimage]# touch Dockerfile  
[root@cs-client myimage]# ls  
app.py Dockerfile  
[root@cs-client myimage]#
```



[27:44](#)

Create a new directory

```
mkdir myimage  
cd myimage
```

Create a Dockerfile

```
FROM python:3.8  
WORKDIR /app  
COPY . .  
CMD ["python", "app.py"]  
~
```



[29:25](#)

```
[root@cs-client myimage]#  
[root@cs-client myimage]# touch Dockerfile  
[root@cs-client myimage]# ls  
app.py  Dockerfile  
[root@cs-client myimage]# vi Dockerfile  
[root@cs-client myimage]# cat Dockerfile  
FROM python:3.8  
WORKDIR /app  
COPY . .  
CMD ["python", "app.py"]  
[root@cs-client myimage]# █
```



[29:30](#)

To build an image

```
docker build -t myimage:1.1 .
```



[30:00](#)

```
[root@cs-client myimage]#  
[root@cs-client myimage]# pwd  
/root/myimage  
[root@cs-client myimage]# ls  
app.py  Dockerfile  
[root@cs-client myimage]#
```



[30:26](#)

```
[root@cs-client myimage]#  
[root@cs-client myimage]# pwd  
/root/myimage  
[root@cs-client myimage]# ls  
app.py  Dockerfile  
[root@cs-client myimage]#  
[root@cs-client myimage]# docker build -t myimage:1.0
```



[31:05](#)

```
[root@cs-client myimage]#  
[root@cs-client myimage]# docker images  
REPOSITORY          TAG          IMAGE ID       CREATED        SI  
ZE  
localhost/myimage   1.0          43ddb7128e51   21 seconds ago 92  
8 MB  
docker.io/library/nginx  latest      6efc10a0510f   6 days ago    14  
7 MB  
docker.io/library/python 3.8         8ada2258068b   6 days ago    92  
8 MB  
[root@cs-client myimage]#
```



[31:31](#)

```
[root@cs-client myimage]#  
[root@cs-client myimage]# docker images  
REPOSITORY          TAG          IMAGE ID        CREATED          SI  
ZE  
localhost/myimage    1.0          43ddb7128e51    21 seconds ago  92  
8 MB  
docker.io/library/nginx    latest       6efc10a0510f    6 days ago      14  
7 MB  
docker.io/library/python    3.8          8ada2258068b    6 days ago      92  
8 MB  
[root@cs-client myimage]#  
[root@cs-client myimage]# docker run localhost/myimage
```



[31:49](#)

```
[root@cs-client myimage]#  
[root@cs-client myimage]# docker run localhost/myimage:1.0  
Hey Buddy!  
[root@cs-client myimage]#
```



[32:18](#)



```
print ("Please Like this Vide👍")
```

2 2 2 2 2 2 2 2 2 2 2 2

32:31

```
[root@cs-client myimage]#  
[root@cs-client myimage]# docker run localhost/myimage:1.0  
Hey Buddy!  
[root@cs-client myimage]# vi app.py  
[root@cs-client myimage]#  
[root@cs-client myimage]# docker build -t myimage:1.1 .
```



32:46

```
[root@cs-client myimage]#
[root@cs-client myimage]# docker run localhost/myimage:1.0
Hey Buddy!
[root@cs-client myimage]# vi app.py
[root@cs-client myimage]#
[root@cs-client myimage]# docker build -t myimage:1.1 .
STEP 1/4: FROM python:3.8
STEP 2/4: WORKDIR /app
--> Using cache fd668160cfb1628cca98110c097be979fbe7a19353f3cd9f39699d
0a59c31ec8
--> fd668160cfb
STEP 3/4: COPY . .
--> ce48d19c902
STEP 4/4: CMD ["python", "app.py"]
COMMIT myimage:1.1
--> ba4b0550e93
Successfully tagged localhost/myimage:1.1
ba4b0550e9315dd2ce101eb2c05e4e3b86014685c21085096acc5de5d501
[root@cs-client myimage]#
```



[32:54](#)

```
[root@cs-client myimage]#
[root@cs-client myimage]# docker images
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
localhost/myimage   1.1         ba4b0550e931     7 seconds ago    928 MB
localhost/myimage   1.0         43ddb7128e51     2 minutes ago    928 MB
docker.io/library/nginx latest        6efc10a0510f     6 days ago       147 MB
docker.io/library/python 3.8         8ada2258068b     6 days ago       928 MB
[root@cs-client myimage]#
```



[33:05](#)

[33:05](#)

```
[root@cs-client myimage]#
[root@cs-client myimage]# docker images
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
localhost/myimage    1.1          ba4b0550e931     7 seconds ago    928 MB
localhost/myimage    1.0          43ddb7128e51     2 minutes ago    928 MB
docker.io/library/nginx latest       6efc10a0510f     6 days ago       147 MB
docker.io/library/python 3.8         8ada2258068b     6 days ago       928 MB
[root@cs-client myimage]#
[root@cs-client myimage]# docker run localhost/myimage:1.1
```



[33:10](#)

```
[root@cs-client myimage]#
[root@cs-client myimage]# docker images
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
localhost/myimage    1.1          ba4b0550e931     7 seconds ago    928 MB
localhost/myimage    1.0          43ddb7128e51     2 minutes ago    928 MB
docker.io/library/nginx latest       6efc10a0510f     6 days ago       147 MB
docker.io/library/python 3.8         8ada2258068b     6 days ago       928 MB
[root@cs-client myimage]#
[root@cs-client myimage]# docker run localhost/myimage:1.1
Please Like this Video
[root@cs-client myimage]#
```



```
[root@cs-client myimage]#  
[root@cs-client myimage]# docker images  
REPOSITORY          TAG          IMAGE ID        CREATED          SIZE  
localhost/myimage    1.1          ba4b0550e931   7 seconds ago   928 MB  
localhost/myimage    1.0          43ddb7128e51   2 minutes ago   928 MB  
docker.io/library/nginx latest        6efc10a0510f   6 days ago      147 MB  
docker.io/library/python 3.8          8ada2258068b   6 days ago      928 MB  
[root@cs-client myimage]#  
[root@cs-client myimage]# docker run localhost/myimage:1.1
```



```
[root@cs-client myimage]#  
[root@cs-client myimage]# docker images  
REPOSITORY          TAG          IMAGE ID        CREATED          SIZE  
localhost/myimage    1.1          ba4b0550e931   7 seconds ago   928 MB  
localhost/myimage    1.0          43ddb7128e51   2 minutes ago   928 MB  
docker.io/library/nginx latest        6efc10a0510f   6 days ago      147 MB  
docker.io/library/python 3.8          8ada2258068b   6 days ago      928 MB  
[root@cs-client myimage]#  
[root@cs-client myimage]# docker run localhost/myimage:1.1
```



```
[root@cs-client myimage]#
[root@cs-client myimage]# docker images
REPOSITORY          TAG          IMAGE ID        CREATED         SIZE
localhost/myimage    1.1          ba4b0550e931    7 seconds ago   928 MB
localhost/myimage    1.0          43ddb7128e51    2 minutes ago   928 MB
docker.io/library/nginx latest        6efc10a0510f    6 days ago      147 MB
docker.io/library/python 3.8          8ada2258068b    6 days ago      928 MB
[root@cs-client myimage]#
[root@cs-client myimage]# docker run localhost/myimage:1.1
```



[30:36](#)

```
[root@cs-client myimage]#
[root@cs-client myimage]# pwd
/root/myimage
[root@cs-client myimage]# ls
app.py  Dockerfile
[root@cs-client myimage]#
[root@cs-client myimage]# docker build -t myimage:1.0 .
STEP 1/4: FROM python:3.8
Resolved "python" as an alias (/etc/containers/registries.conf.d/000-shortnames.conf)
Trying to pull docker.io/library/python:3.8...
█
```





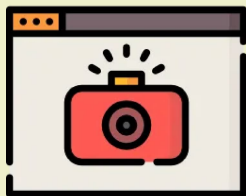
23:25

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